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Background

California Wildfire Crisis

- Increasing frequency and intensity
- Urgent need for effective evacuation strategies

Evacuation Challenges

- Unpredictable fire spread
- Large-scale evacuations overwhelm infrastructure

Our Approach

- Combine ABMs with RL
- Simulate individual behaviors and adaptive decision-making
- Aim: Develop more effective, real time evacuation strategies



Figure 1 Sacramento, CA Wildfire Map and Climate Risk Report

Materials and Methods

Software tools

- NetLogo + Python for ABM Implementation
- ArcGIS for geospatial modeling and visualization

RL Integration

- Agents adapted strategies dynamically in response to wildfire.

Simulation Process

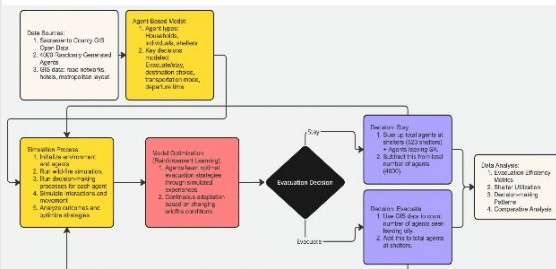


Figure 2 Flowchart explaining the methods and models

Results

Evacuation Timing

- Mean evacuation time: ~25 minutes after wildfire start
- Bimodal distribution suggests two main waves of evacuation
- Most evacuations occurred between 15-45 minutes after wildfire onset.

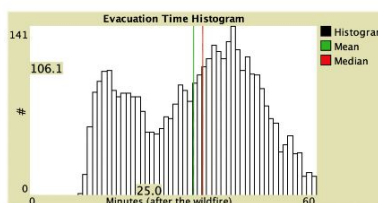


Figure 3 Histogram displaying evacuation times during the simulation

Evacuation Success

- ~90% of total population evacuated by the end
- Cars evacuated faster at the start, but pedestrians caught up by ~75 minutes.
- At ~45 minutes, approximately 50% of the population had evacuated.

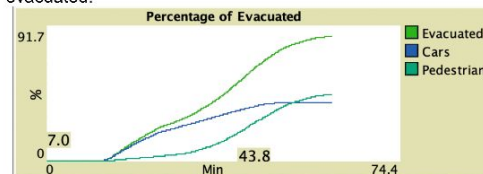


Figure 4 Histogram displaying evacuation times during the simulation

Measuring Casualties

- ABM consistently simulated fewer casualties
- More consistency between Agent-Based Modeling

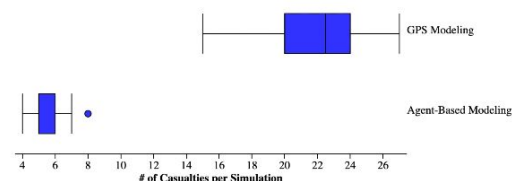


Figure 5 Boxplot displaying differences in casualties between GPS & ABM modeling

Conclusions and Future Work

Key Findings

- Through Agent-Based Modeling, we can see a significant reduction in casualties compared to traditional GPS modeling.
- The ABM achieved a 90% evacuation rate, indicating a higher success rate in agents reaching safety.
- The ABM's ability to continuously adapt to changing wildfire conditions significantly improved evacuation outcomes.

Recommendations for Future Researchers:

- Focus on incorporating real-time data sources, such as live GPS tracking and emergency broadcasts, to further enhance the model's responsiveness and accuracy.
- Adapt and test ABM and RL for other disaster types like floods, hurricanes, and earthquakes to assess its versatility and effectiveness across different emergencies.
- Conduct further analysis on the impact of socioeconomics and demographic factors on evacuation behaviors. Understanding how different populations respond to evacuation orders can help further optimize strategies for diverse communities.

Major Citations

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